

SITE-PROBE ACCURACY IS NOT A HARMONIZATION SCORE: AN AUDIT OF EVALUATION PRACTICE FOR MULTI-SITE MRI

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ABSTRACT

Image-level MRI harmonization is often scored by how poorly a frozen site classifier recognizes the acquisition site of harmonized images. We audit this practice on raw ABIDE I T1-weighted MRI (1,112 subjects, 17 sites), using ten outputs from seven harmonization families as test cases and paired subject-level intervals. (i) Verdicts depend on how the probe is trained: five probes trained with the benchmark’s recipe gave the same CycleGAN outputs site balanced accuracies of 0.13–0.59; five converged probes agreed (0.44–0.54) but found more site information in eight of ten outputs, e.g. 0.70 versus 0.21 for aggressive StarGAN. (ii) The lowest accuracy, at chance, came from the most destructive output. (iii) Diffusion translation changed intensities least but barely moved them toward the target site. (iv) Probes trained on harmonized outputs recovered the source site from whole heads, skull-stripped brains and brain intensity histograms alike. We propose a reporting checklist.

Index Terms— MRI harmonization, evaluation, site classification, domain shift, reproducibility

1. INTRODUCTION

Acquisition differences between sites confound pooled MRI analyses and degrade models on unseen scanners [1, 2]. Image-level harmonization translates scans toward a common appearance. Travelling subjects [3] or paired protocols [4] allow direct checks but cover few subjects and scanners, so methods applied to large retrospective cohorts are judged indirectly [5]. A common proxy is a *site probe*: a classifier trained to recognize the site on unharmonized images and applied, frozen, to harmonized ones [2, 6]. Lower accuracy is read as removal of site information and is paired with similarity to the input, such as PSNR, and with intensity-distribution distances.

Each proxy measures less than it appears to. A frozen probe reports what one classifier recovers; it can also fail because outputs are unfamiliar, damaged or noisy. Removing

an attribute until a fixed adversary fails does not stop a newly trained classifier from recovering it [7], and for harmonized features site classifiers retrained after ComBat still predicted site above chance [8]. Similarity to the input rewards doing nothing, and a small input-to-output distance says nothing about the target site.

We audit these practices on outputs of seven harmonization families. Contributions: (1) a subject-level protocol with paired uncertainty, valid probe controls, separate reporting of target-site subjects, and target alignment measured apart from change; (2) evidence that probe verdicts depend on how the probe was trained, that the lowest site accuracy can signal destruction, and that small change does not imply alignment; (3) a test of removed versus hidden site information using probes trained on harmonized images, with geometry-only, intensity-only and brain-only controls; and (4) a reporting checklist.¹ We neither propose nor rank harmonizers.

2. DATA AND AUDITED OUTPUTS

Cohort. Raw ABIDE I T1-weighted scans [9] of 1,112 subjects from 17 sites are split by subject, stratified by site, into 890 training, 113 validation and 109 test subjects. Volumes are reoriented, clipped to foreground percentiles $[p_1, p_{99}]$ and scaled to $[0, 1]$. One axial slice per subject is chosen by foreground area, cropped with a dilated head mask and resized to 256×256 . NYU, the largest site, is the target; 19 test subjects are from NYU and 90 from 16 source sites. Exact ties made slice choice host-dependent, so slices are frozen and verified for every output.

Audited outputs. Earlier benchmark outputs were evaluated without retraining: NeuroCombat [10, 1] per pixel, fit on the test cohort itself (transductive); histogram matching of foreground quantiles to a pooled 17-site training reference [11]; CycleGAN [12]; StarGAN [13], aggressive and conservative; DLEST [14] after 1,000 and 1,500 steps; conditional diffusion image-to-image translation [15, 16] (strength 0.35, 50 DDIM steps); and adapted volumetric HCLD [17].

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¹Code, the evaluation protocol with all amendments, and per-subject results: <https://github.com/muqsitamir/HarmonIt>.

Learned methods were trained on the training split. Test subjects were inspected in the earlier benchmark, so this is a retrospective reanalysis; each audit experiment was written into a protocol before its outcome was seen, recording what had been observed.

3. AUDIT PROTOCOL

Change and target alignment. PSNR on a fixed $[0, 1]$ scale and normalized cross-correlation (XCorr) compare output with input on whole images; generators return NYU inputs almost unchanged (PSNR infinite or above 150 dB), so we summarize the 90 source subjects. *Change* is the per-subject Wasserstein-1 distance W between input and output intensities. *Target alignment* is the Wasserstein-1 distance and KL divergence between foreground intensities (defined on the raw slice) and a reference from the 147 NYU training subjects; ΔW_{NYU} is the output’s distance minus the input’s.

Site probes. Probes are single-channel ResNet-18 networks [18] predicting 17 sites, trained with AdamW (learning rate 3×10^{-4} , batch 64, 500 steps) and random affine augmentation. The *frozen* probe is the earlier benchmark’s checkpoint. *Benchmark-recipe* probes retrain it on random valid training slices (seeds 42, 1–4), selected on validation balanced accuracy (BA) and tested once. *Converged* probes (seeds 5–9) use four times the steps with cosine learning-rate decay and are tested at the final epoch. Shuffled-label controls permute labels between subjects within each split. To separate removed from hidden site information, *slice probes* (benchmark recipe, seeds 1–3) are trained on fixed slices of raw images or of one method’s outputs, for three methods chosen in advance to span intensity, adversarial and diffusion families. Controls restrict the input: slice probes on filled binary head silhouettes (geometry only), and multinomial logistic regression on 50-bin foreground intensity histograms (intensity only), trained on raw slices or on a method’s outputs. A brain-only variant repeats slice and histogram probes after multiplying raw and harmonized slices by the raw volume’s HD-BET brain mask [19], with binary brain masks as its geometry control. The converged, silhouette, histogram and brain-only experiments were added after earlier results and before their own, as recorded in the protocol.

Statistics. Source BA is macro-averaged over the 16 source sites (chance 1/16). Intervals are 95% percentiles of 2,000 bootstrap replicates resampling subjects within site, shared across methods and probes (paired) and conditional on fixed checkpoints.

4. RESULTS

Probe validity. On raw source images the frozen probe reached BA 0.95 [0.90, 1.00], benchmark-recipe probes 0.81–0.98 and converged probes 1.00; shuffled-label controls scored 0.03. Probes learn site-related structure, not

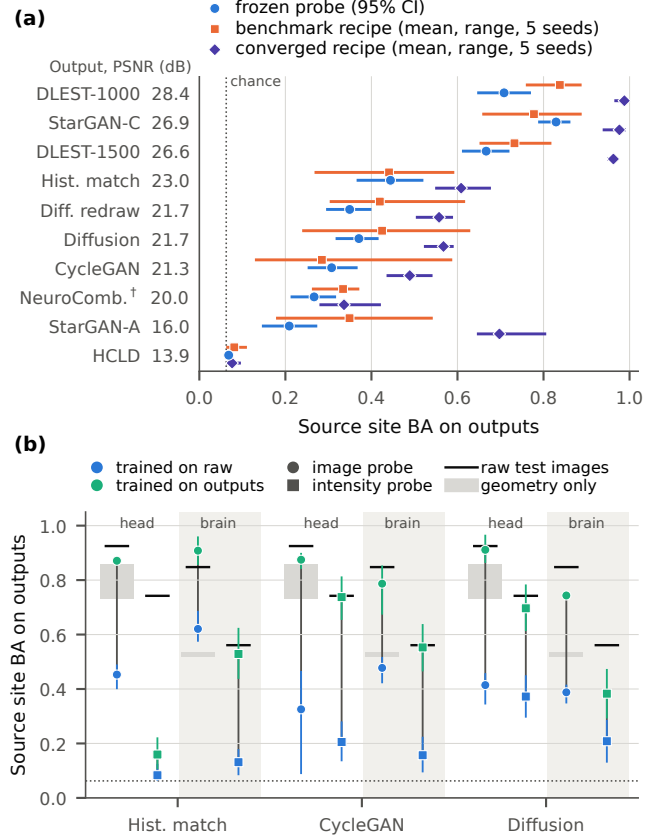


Fig. 1. (a) Source site BA of each output, ordered by PSNR. (b) Probes trained on raw slices (blue) or on a method’s outputs (aqua), tested on that method’s outputs, on whole heads and brain-only slices: image probes (circles, mean and range over 3 seeds) and intensity-histogram probes (squares, 95% interval). Black bar: raw-trained probe on raw test slices. Grey: image probes given only head silhouettes or brain masks.

necessarily scanner appearance: head silhouettes alone gave 0.73–0.86.

Verdicts depend on probe training. Benchmark-recipe probes assigned very different site BA to the same outputs (Table 1, Fig. 1a): CycleGAN 0.13–0.59, diffusion 0.24–0.63, histogram matching 0.27–0.59 and aggressive StarGAN 0.18–0.54, with Kendall τ 0.51–0.87 between method orderings. Their validation BA fluctuated between epochs (e.g. 0.88, 0.50, 0.87), and raw accuracy did not reveal it: the probe with raw BA 0.98 gave CycleGAN 0.20. Converged probes (validation BA within 0.035 over the last five epochs) agreed (τ 0.82–0.96; per-output ranges at most 0.16, within the widest single-probe interval of 0.18), but every converged probe exceeded the frozen probe’s interval on all outputs except HCLD and NeuroCombat: CycleGAN 0.44–0.54 (frozen 0.31), diffusion 0.52–0.59 (0.37), aggressive StarGAN 0.65–0.81 (0.21) and DLEST 0.95–1.00 (0.67–0.71). Orderings

Table 1. Source-subject results ($n = 90$). PSNR and XCorr compare output with input; ΔW_{NYU} : change in Wasserstein-1 distance to the NYU reference (negative is closer). Site BA: frozen probe with 95% interval; five benchmark-recipe and five converged probes as mean [min, max]. Chance is 0.06. [†]Fit on the test cohort. [‡]Pooled 17-site reference.

Method	Change and target alignment			Source-site balanced accuracy of probes trained on raw images		
	PSNR	XCorr	ΔW_{NYU}	Frozen probe	Benchmark recipe	Converged recipe
Raw input	—	—	0	0.95	0.90 [0.81, 0.98]	1.00 [1.00, 1.00]
NeuroCombat [†]	20.0	0.940	+0.003	0.27 [0.21, 0.32]	0.33 [0.26, 0.37]	0.34 [0.28, 0.42]
Histogram matching [‡]	23.0	0.982	-0.053	0.44 [0.37, 0.52]	0.44 [0.27, 0.59]	0.61 [0.55, 0.68]
CycleGAN	21.3	0.977	-0.036	0.31 [0.25, 0.37]	0.28 [0.13, 0.59]	0.49 [0.44, 0.54]
StarGAN (aggr.)	16.0	0.896	-0.019	0.21 [0.15, 0.27]	0.35 [0.18, 0.54]	0.70 [0.65, 0.81]
StarGAN (cons.)	26.9	0.991	-0.021	0.83 [0.79, 0.86]	0.78 [0.66, 0.89]	0.98 [0.94, 0.99]
DLEST 1500	26.6	0.993	-0.020	0.67 [0.61, 0.72]	0.73 [0.65, 0.82]	0.96 [0.95, 0.97]
DLEST 1000	28.4	0.993	-0.009	0.71 [0.65, 0.77]	0.84 [0.76, 0.89]	0.99 [0.96, 1.00]
Diffusion img2img	21.7	0.948	-0.001	0.37 [0.32, 0.42]	0.42 [0.24, 0.63]	0.57 [0.52, 0.59]
second draw	21.7	0.948	-0.002	0.35 [0.29, 0.40]	0.42 [0.30, 0.62]	0.56 [0.50, 0.59]
Adapted HCLD	13.9	0.706	+0.108	0.07 [0.06, 0.08]	0.08 [0.06, 0.11]	0.08 [0.06, 0.10]

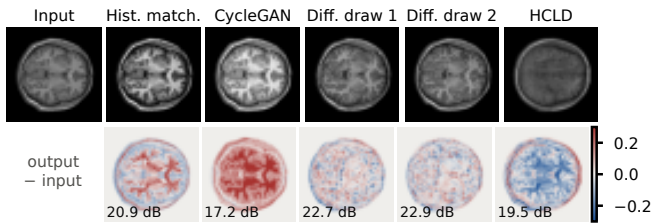


Fig. 2. Source test subject with median PSNR over all outputs. Bottom: output minus input, with PSNR. HCLD loses anatomy; diffusion adds texture that differs between draws.

changed (τ with the frozen probe 0.64–0.73): the frozen probe placed aggressive StarGAN second lowest, converged probes above histogram matching. A probe that has not converged overstates harmonization.

The lowest site accuracy came from destruction. HCLD reached BA 0.07 [0.06, 0.08] under the frozen probe and 0.06–0.11 under every retrained probe, at chance, with the lowest PSNR (13.9 dB) and XCorr (0.71), visible loss of anatomy (Fig. 2), and intensities moved away from the target ($\Delta W_{\text{NYU}} = +0.108$).

Small change is not target alignment. Diffusion translation changed intensities least ($W = 0.008$) yet moved them barely closer to NYU ($\Delta W_{\text{NYU}} = -0.0013$ [−0.0023, −0.0004], 1% of the raw distance) and raised the KL divergence to NYU from 0.30 to 0.83, whereas histogram matching (−0.053) and CycleGAN (−0.036) moved toward the target. Its outputs are also a random draw: resampling the same checkpoint gave images differing from the first draw by a mean foreground absolute difference of 0.085, more than from the input (0.075), at unchanged PSNR (21.71 vs. 21.73 dB).

Removed or hidden? Slice probes trained on raw slices reached BA 0.92 on raw test slices (shuffled-label control 0.07 [0.03, 0.10]) but 0.45, 0.33 and 0.41 on histogram-matched,

CycleGAN and diffusion outputs, the drop usually reported as harmonization. Probes trained on each method’s outputs recovered the source site from its test outputs with BA 0.87, 0.87 and 0.91 (Fig. 1b), with disjoint intervals for every seed. Because head geometry identifies sites, we repeated this on brain-only slices: raw-trained probes reached 0.85 on raw brains but 0.62, 0.48 and 0.39 on the outputs, and probes trained on outputs 0.91, 0.79 and 0.74, again disjoint for every seed with best-validation checkpoints (five of nine with final-epoch ones); brain masks alone gave 0.52–0.54. Intensity histograms remove geometry. On whole heads, a histogram probe trained on outputs recovered little from histogram matching (0.16 [0.11, 0.22]), whose head histograms are matched by construction, but recovered CycleGAN (0.74) and diffusion (0.70) as well as raw slices (0.74). Restricted to the brain it recovered all three: 0.53 [0.44, 0.63], 0.55 [0.47, 0.64] and 0.38 [0.29, 0.47], against 0.56 on raw brains and 0.13–0.21 for the raw-trained probe. The methods transformed the site signature rather than removing it.

5. DISCUSSION AND RECOMMENDATIONS

Low site accuracy from a frozen probe arose from an unconverged probe, from destruction, and, for every method we could test, from site information that probes trained on outputs recovered, also within the brain and from intensities alone; small change from the input was compatible with almost no movement toward the target. Site labels are also a confounded target: head and brain geometry predict them, so image-probe accuracy cannot certify scanner-effect removal. We recommend that evaluations (1) report several independently trained probes with their recipe and evidence of convergence, including probes trained on harmonized outputs; (2) add probes restricted to the channel a method should change, such as brain intensities, with geometry-only controls; (3) report each probe’s raw accuracy and a subject-level shuffled-label control; (4) measure alignment to a held-out

target reference separately from change, excluding target-site subjects from source summaries; (5) compute paired uncertainty over subjects, not pixels or slices; (6) evaluate several draws of stochastic generators; and (7) read low site accuracy together with fidelity.

Limitations. We use one cohort, one target site and one axial slice per subject, so volumetric effects are not assessed. Without anatomical ground truth, PSNR and XCorr measure agreement with the input, not correctness. Brain masks remove the head silhouette but not brain shape, slice level or orientation, and site is confounded with demographics; our controls bound these channels but do not separate scanner effects from biological site differences. Slice probes use the benchmark recipe, generators were not retuned, and one baseline is transductive.

6. COMPLIANCE WITH ETHICAL STANDARDS

This retrospective study used open-access human data from ABIDE I [9]; ethical approval was not required, as confirmed by the data license.

7. ACKNOWLEDGMENTS

[Funding.] The authors have no relevant financial or non-financial interests to disclose. Generative AI tools (OpenAI Codex, Anthropic Claude) assisted with analysis code and manuscript drafting; the authors verified all code, results and text and take full responsibility for them.

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